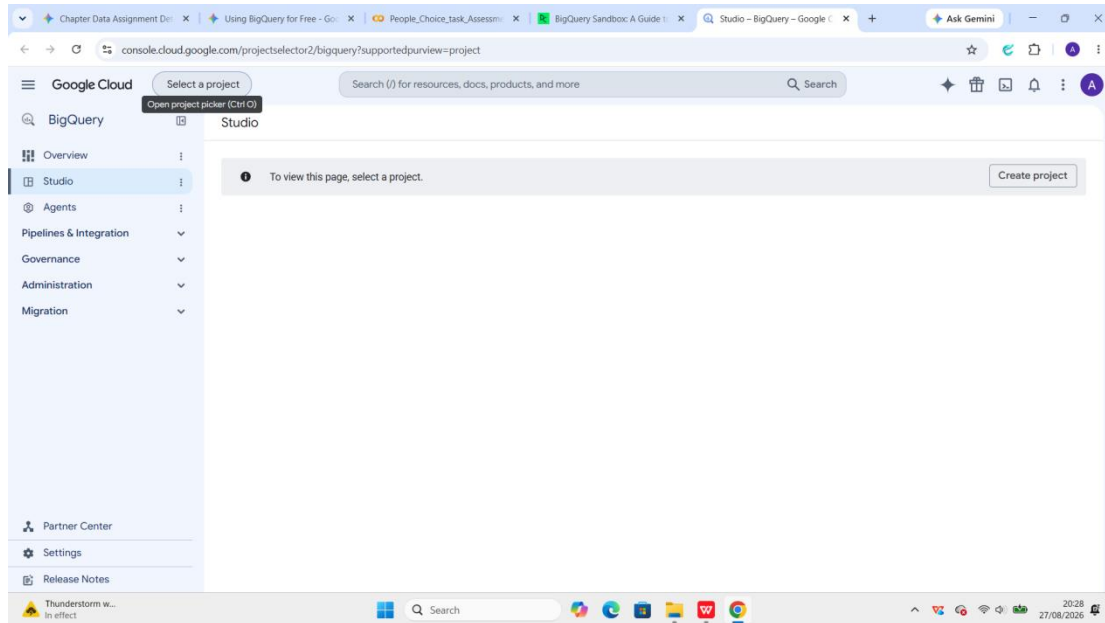
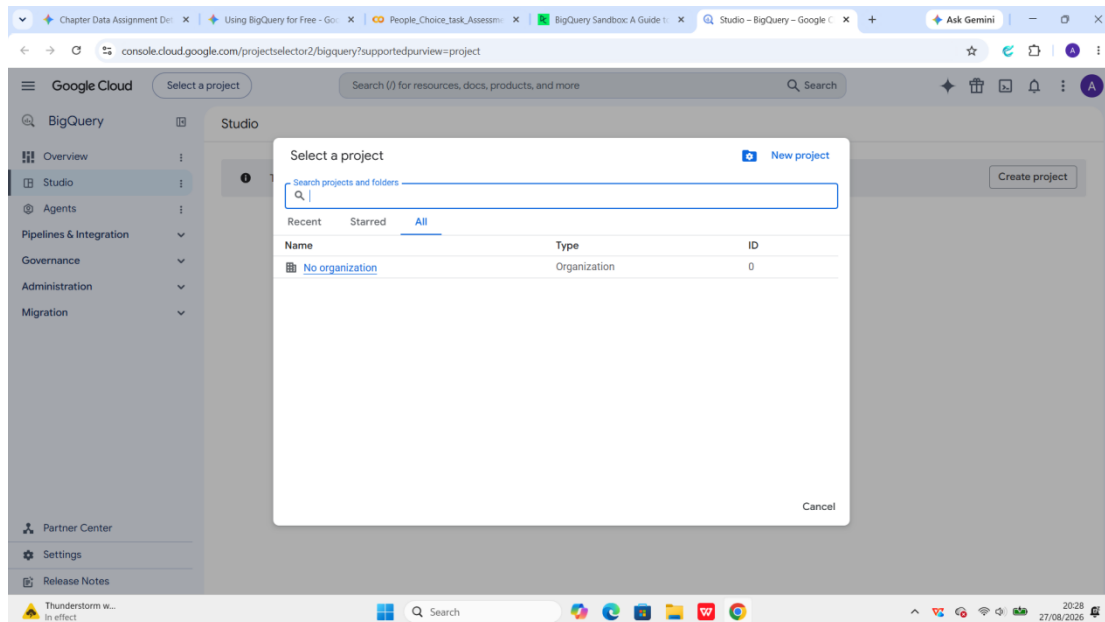


Q 1.) How to create project_id in bigquery UI?

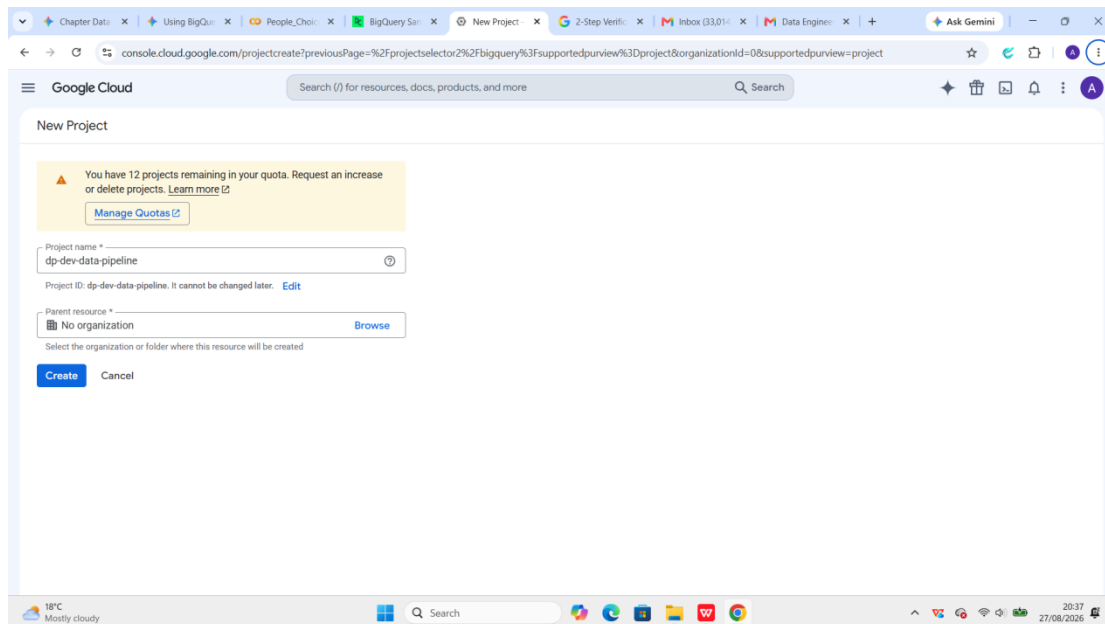
Step 1:- Select a Project



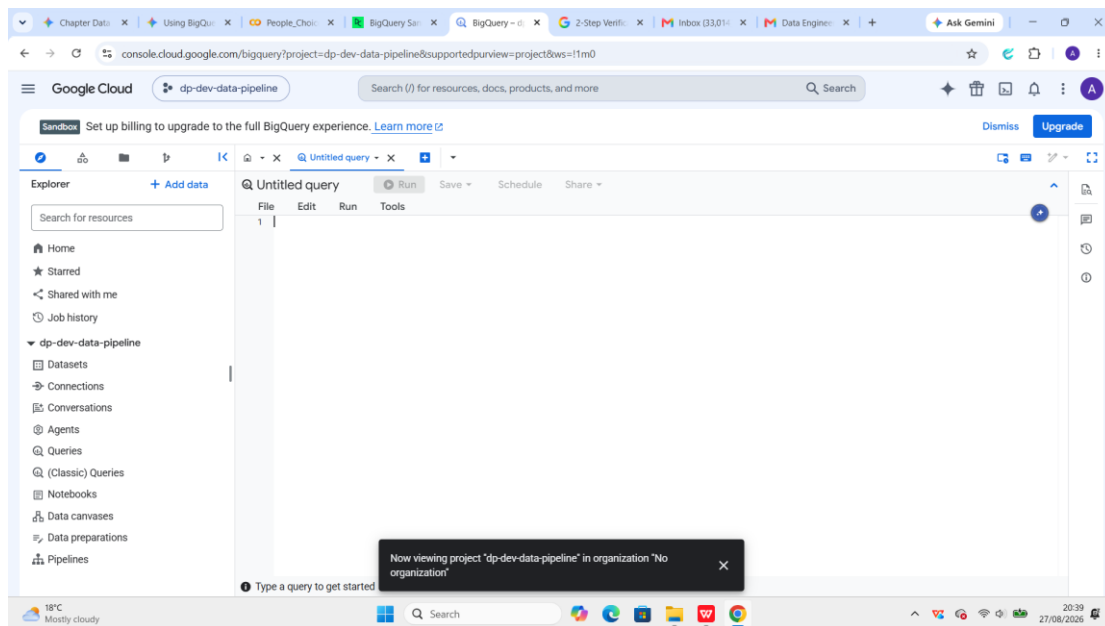
Step 2:- Click on New Project and give name (dp-dev-data-pipeline)



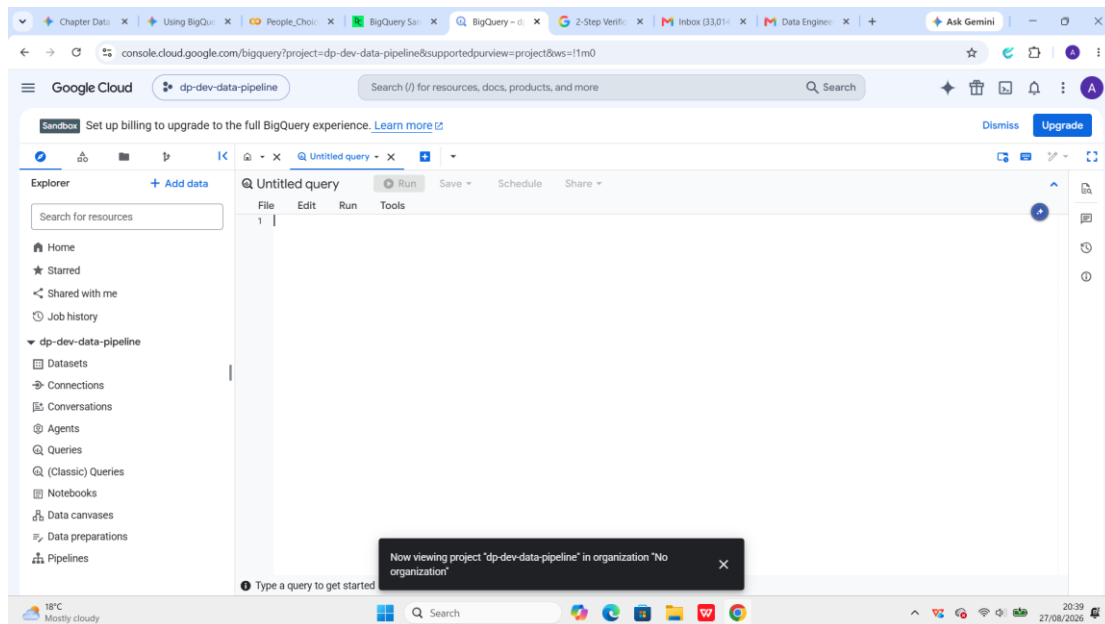
Step 3:- Do not create on any billing



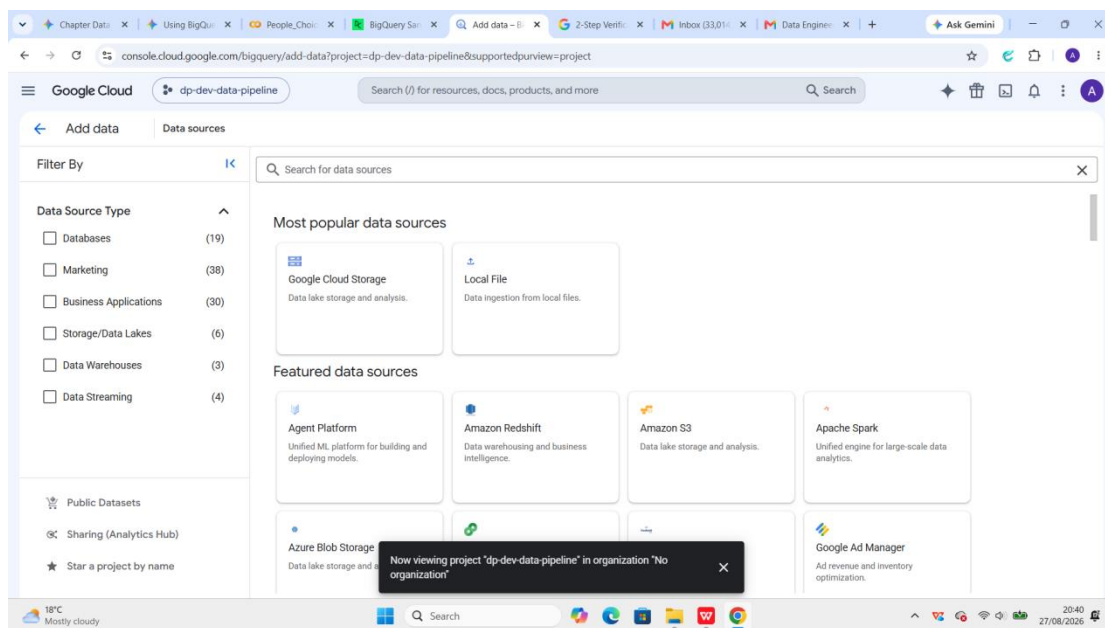
Step 4:- Once project is create open bigquery from left menu or direct search

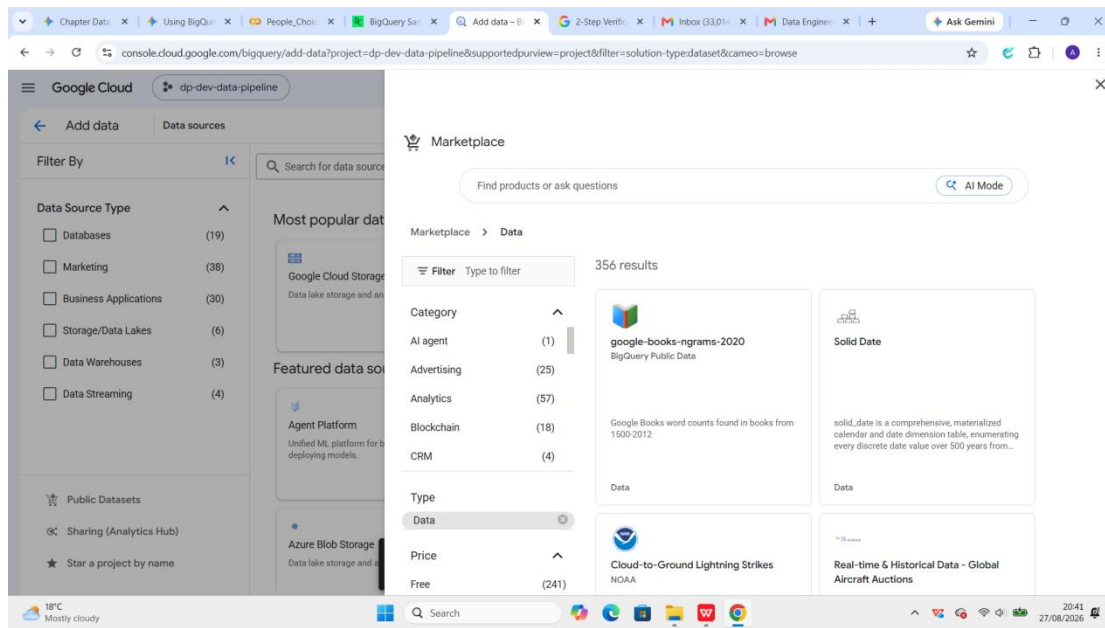


Step 5:- Click on Add data

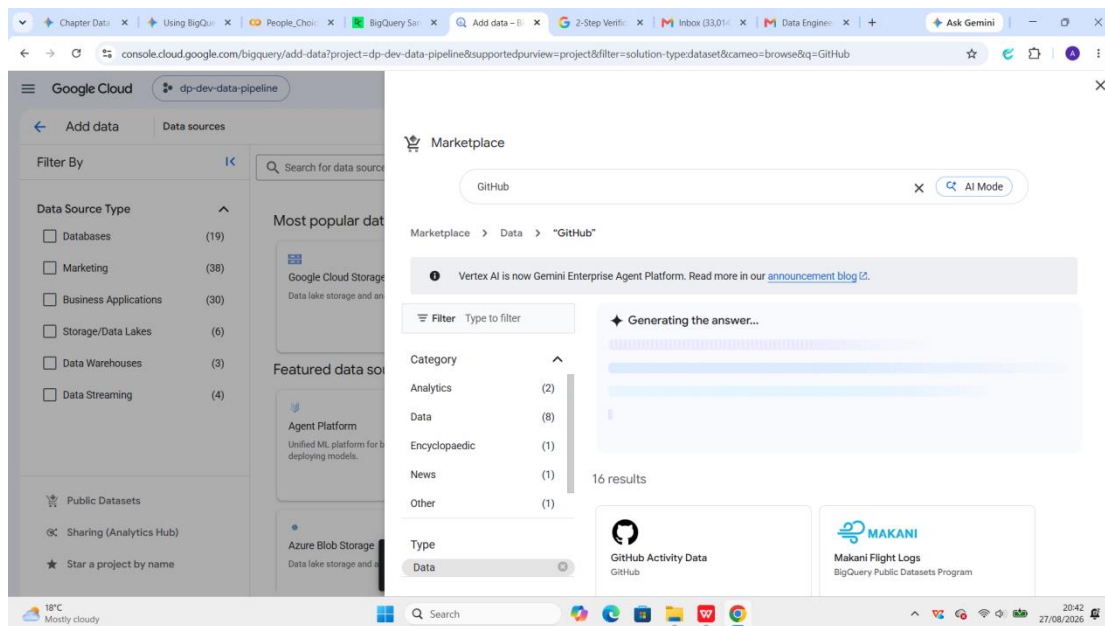


Step 6:- Click on Public Datasets





Step 7:- Type GitHub in space bar



Step 8:- Click on View Dataset

Google Cloud console interface showing the 'GitHub Activity Data' dataset page. The page includes a sidebar with 'Add data' and 'Data sources' options, a search bar, and a main content area with 'Overview' and 'Additional details' sections. The 'Overview' section describes the dataset as a snapshot of GitHub activity from over 3M open source repositories. The 'Additional details' section lists the type as 'Data', category as 'Encyclopedic, Other', dataset source as 'GitHub', cloud service as 'BigQuery', and expected update frequency as 'Weekly'.

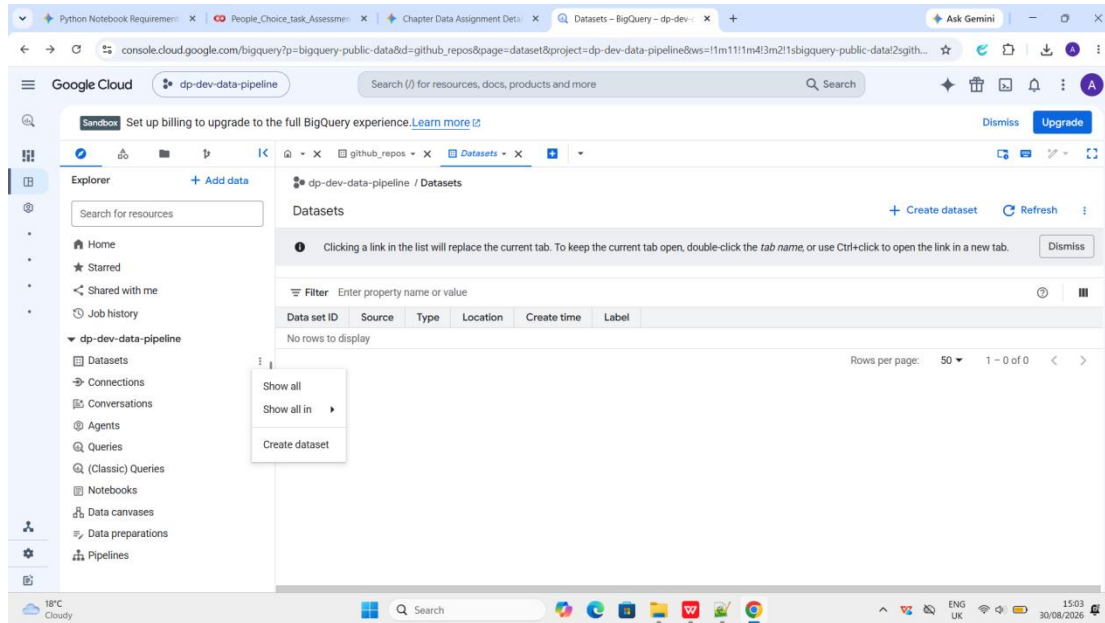
Step 9:- Click on the STAR button

Google Cloud console interface showing the 'github_repos' dataset page. The page includes a sidebar with 'Add data' and 'Data sources' options, a search bar, and a main content area with 'Overview' and 'Details' sections. The 'Overview' section shows a table of dataset tables with columns: Table ID, Type, Create time, Expiry time, and Label. The 'Details' section shows the 'github_repos' dataset with a 'STAR' button.

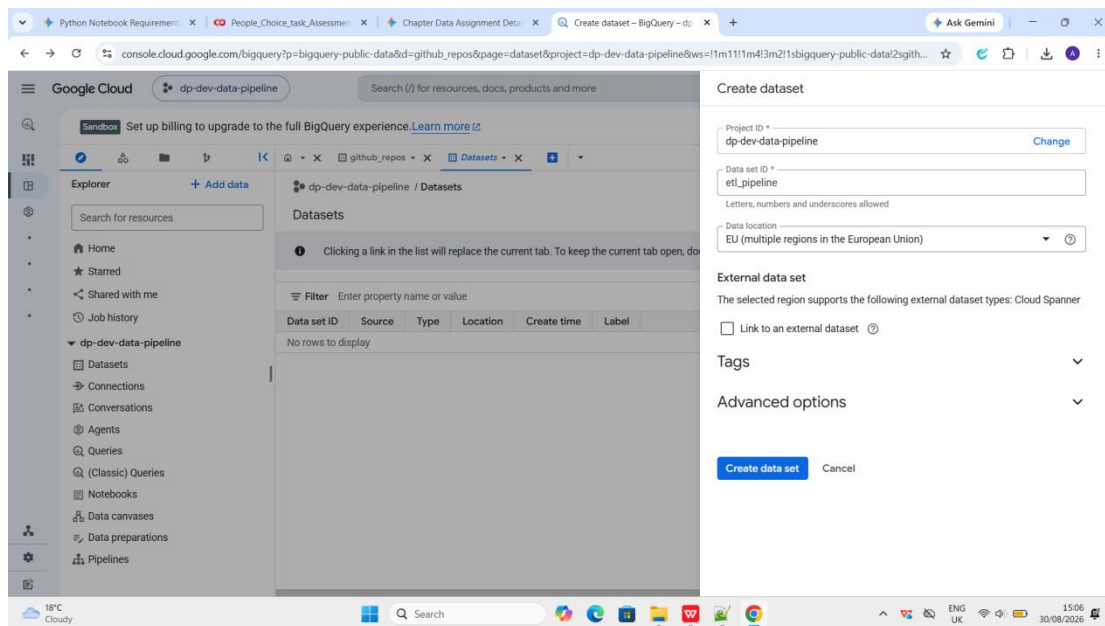
Table ID	Type	Create time	Expiry time	Label
commits	Table	23 Mar 20...	None	dataplex-data-documentation-published-location: us-central1
contents	Table	12 Mar 20...	None	None
files	Table	12 Mar 20...	None	None
languages	Table	16 Mar 20...	None	None
licenses	Table	13 Mar 20...	None	None
sample_commits	Table	23 Jun 201...	None	None
sample_contents	Table	24 May 20...	None	None
sample_files	Table	24 May 20...	None	None
sample_repos	Table	24 May 20...	None	None

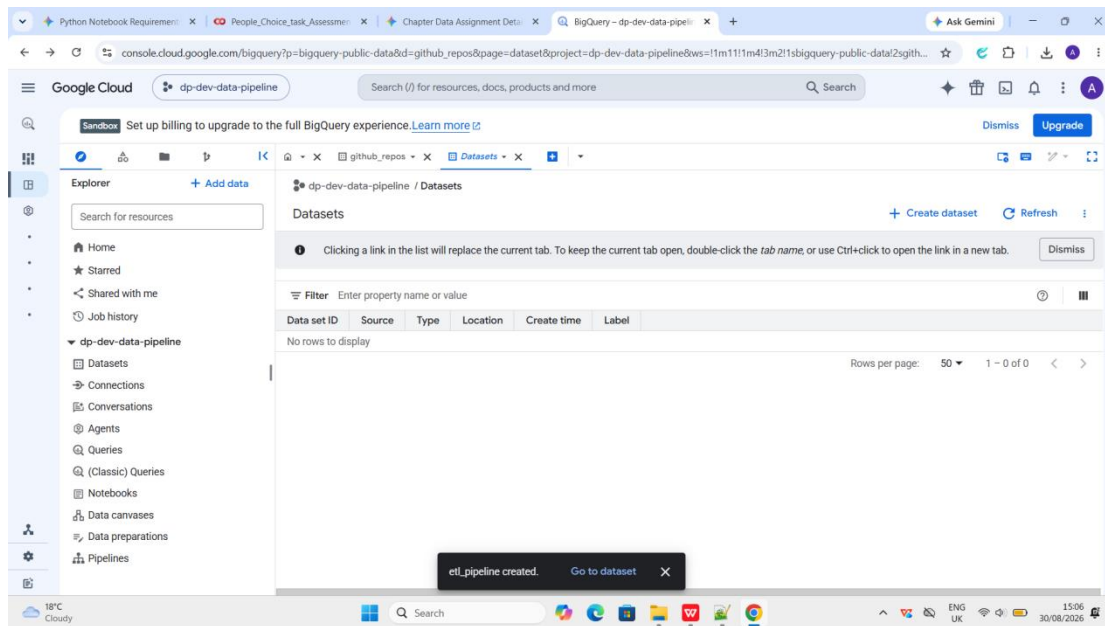
Q 2.) How to create dataset_id in BigQuery UI?

Step 1:- Click on Add data at left side of the bigquery UI.

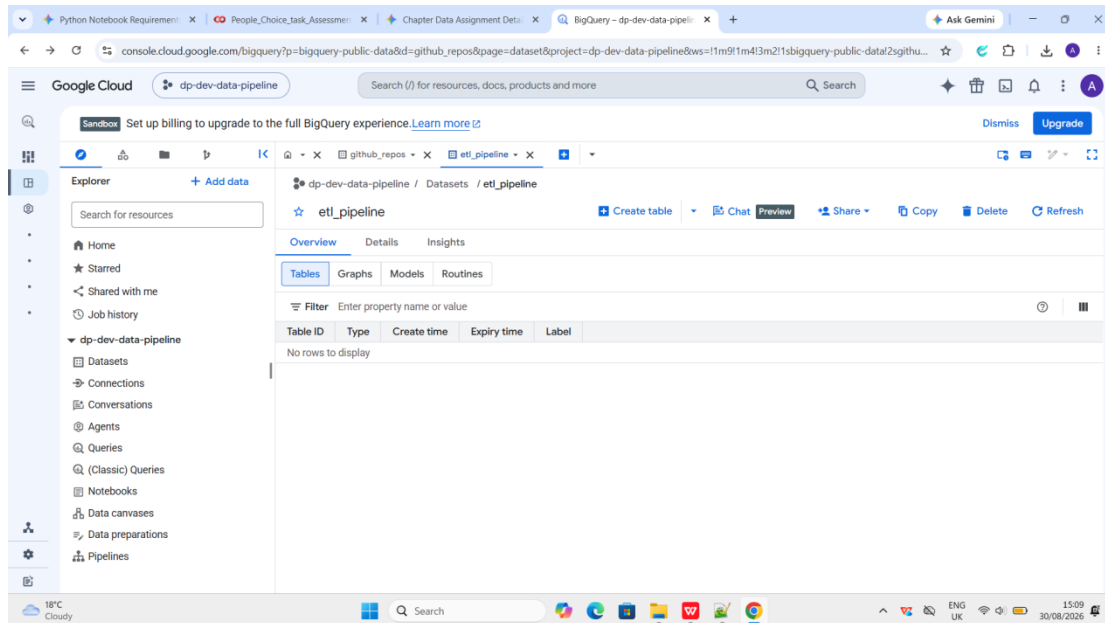


Step 2:- Give the name of dataset id: etl_pipeline





Step 3:- Now below is the UI of the dataset (etl_pipeline) inside project_id (dp-dev-data-pipeline)



Step 4:- Reload the BigQuery UI console I see the table (university_chapters) got created.

The screenshot shows the Google Cloud BigQuery console interface. The breadcrumb navigation indicates the path: `dp-dev-data-pipeline / Datasets / etl_pipeline`. The table `university_chapters` is listed with the following details:

Table ID	Type	Create time	Expiry time	Label
<code>university_chapters</code>	Table	30 Aug 2026, 15:14:30 UTC+1	29 Oct 2026, 14:14:30 UTC	None

The screenshot shows the 'Schema' tab for the `university_chapters` table. The schema details are as follows:

Field name	Type	Mode	Description	Key	Collation	Default value	Data governance tags	Policy tags
<code>chapter_id</code>	STRING	REQUIRED	-	-	-	-	-	-
<code>chapter_name</code>	STRING	REQUIRED	-	-	-	-	-	-
<code>city</code>	STRING	NULLABLE	-	-	-	-	-	-
<code>state</code>	STRING	NULLABLE	-	-	-	-	-	-
<code>longitude</code>	FLOAT	REQUIRED	-	-	-	-	-	-
<code>latitude</code>	FLOAT	REQUIRED	-	-	-	-	-	-

Python Notebook Requirements | People Choice Task Assessment | Chapter Data Assignment Details | etl_pipeline - BigQuery - dp-dev-data-pipeline | Ask Gemini

console.cloud.google.com/bigquery?p=bigquery-public-data&id=github_repos&page=dataset&project=dp-dev-data-pipeline&ws=1m161m43m211sbigquery-public-data2sgith...

Google Cloud dp-dev-data-pipeline Search (/) for resources, docs, products and more

Sandbox Set up billing to upgrade to the full BigQuery experience. [Learn more](#) Dismiss Upgrade

Explorer + Add data

Search for resources

Home

Starred

Shared with me

Job history

dp-dev-data-pipeline

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

dp-dev-data-pipeline / Datasets / etl_pipeline / Tables / university_chapters

☆ university_cha... Query Chat Open In + Share Copy Snapshot Delete Refresh

Schema Details Preview Insights Lineage Data profile Data Quality

Table info Edit details

Table ID dp-dev-data-pipeline.etl_pipeline.university_chapters

Created 30 Aug 2026, 15:14:30 UTC+1

Last modified 30 Aug 2026, 15:14:30 UTC+1

Table expiry 29 Oct 2026, 14:14:30 UTC

Data location EU

Default collation

Default rounding mode ROUNDING_MODE_UNSPECIFIED

Case insensitive false

Description

Labels

Primary key(s)

Tags

Storage info

Number of rows 91

18°C Cloudy Search

Python Notebook Requirements | People Choice Task Assessment | Chapter Data Assignment Details | etl_pipeline - BigQuery - dp-dev-data-pipeline | Ask Gemini

console.cloud.google.com/bigquery?p=bigquery-public-data&id=github_repos&page=dataset&project=dp-dev-data-pipeline&ws=1m161m43m211sbigquery-public-data2sgith...

Google Cloud dp-dev-data-pipeline Search (/) for resources, docs, products and more

Sandbox Set up billing to upgrade to the full BigQuery experience. [Learn more](#) Dismiss Upgrade

Explorer + Add data

Search for resources

Home

Starred

Shared with me

Job history

dp-dev-data-pipeline

Datasets

Connections

Conversations

Agents

Queries

(Classic) Queries

Notebooks

Data canvases

Data preparations

Pipelines

dp-dev-data-pipeline / Datasets / etl_pipeline / Tables / university_chapters

☆ university_cha... Query Chat Open In + Share Copy Snapshot Delete Refresh

Schema Details Preview Insights Lineage Data profile Data Quality

Row	chapter_id	chapter_name	city	state	longitude	latitude
1	AL0107	University of Alabama	Tuscaloosa	AL	-87.5346091...	33.2065366...
2	AL0114	University of Auburn	Auburn	AL	-85.489820...	32.6021289...
3	AR0124	University of Central Arkansas	Conway	AR	-92.4533186...	35.0873368...
4	AR0110	U of A Razorback	Fayetteville	AR	-94.1609101...	36.076382165
5	AR0130	Arkansas Tech University	Russellville	AR	-93.1368187...	35.2784269...
6	AR-0113	Arkansas State University	Jonesboro	AR	-90.6757361...	35.8416758...
7	AZ0070	University of Arizona	Tucson	AZ	-110.953881...	32.2318967...
8	CA0355	California Polytechnic State Uni...	San Luis Obispo	CA	-120.463191...	35.2743091...
9	CA0300	Chico State University	Chico	CA	-121.835463...	39.7399817...
10	CO0108	Colorado State University	Fort Collins	CO	-105.078300...	40.599236561
11	DE0016	University of Delaware	Newark	DE	-75.7580372...	39.6791093...
12	FL0110	Florida State University	Tallahassee	FL	-84.3042726...	30.438110943
13	FL-0126	University of Florida/Gainesville	Gainesville	FL	-82.3552443...	29.6423628...
14	GA0020	Abraham Baldwin Agricultural C...	Tifton	GA	-83.5100644...	31.4634182...
15	GA0133	University of Georgia	Athens	GA	-83.3832464...	33.955464353

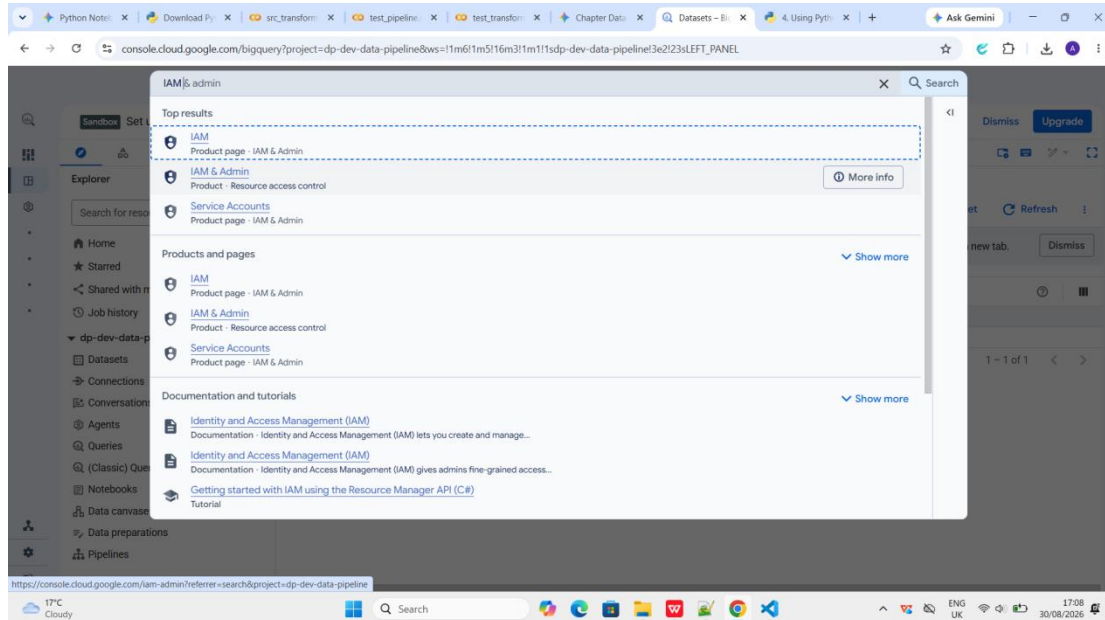
Results per page: 50 1 - 50 of 91

18°C Cloudy Search

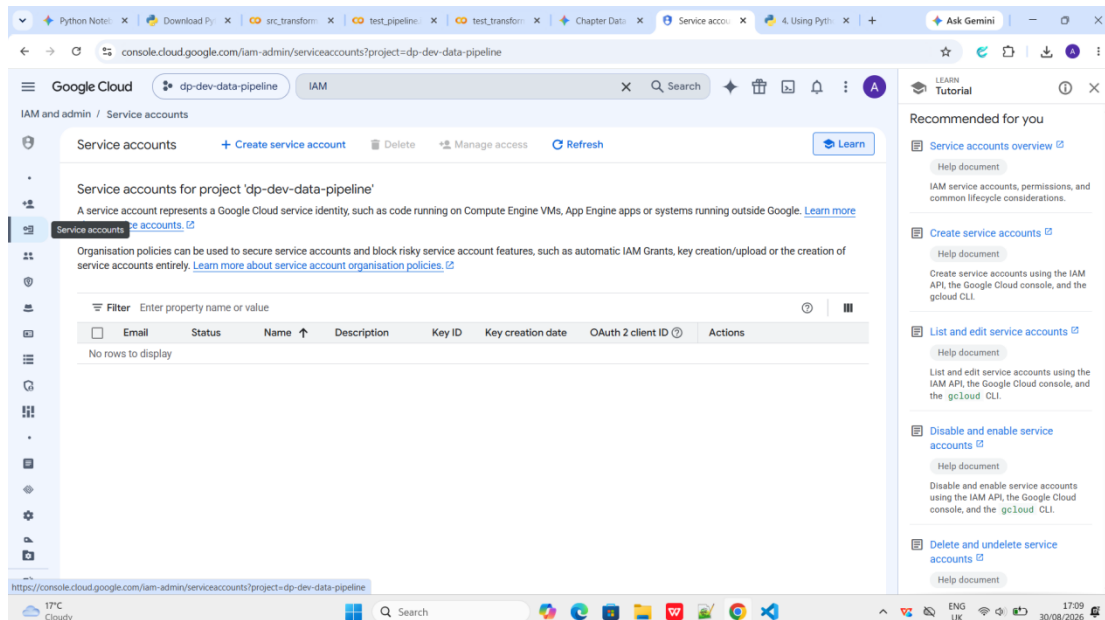
Q 3.) How to create and set up a Google Cloud service account key file?

Step 1:- Navigate to Service Accounts in the left hand navigation menu

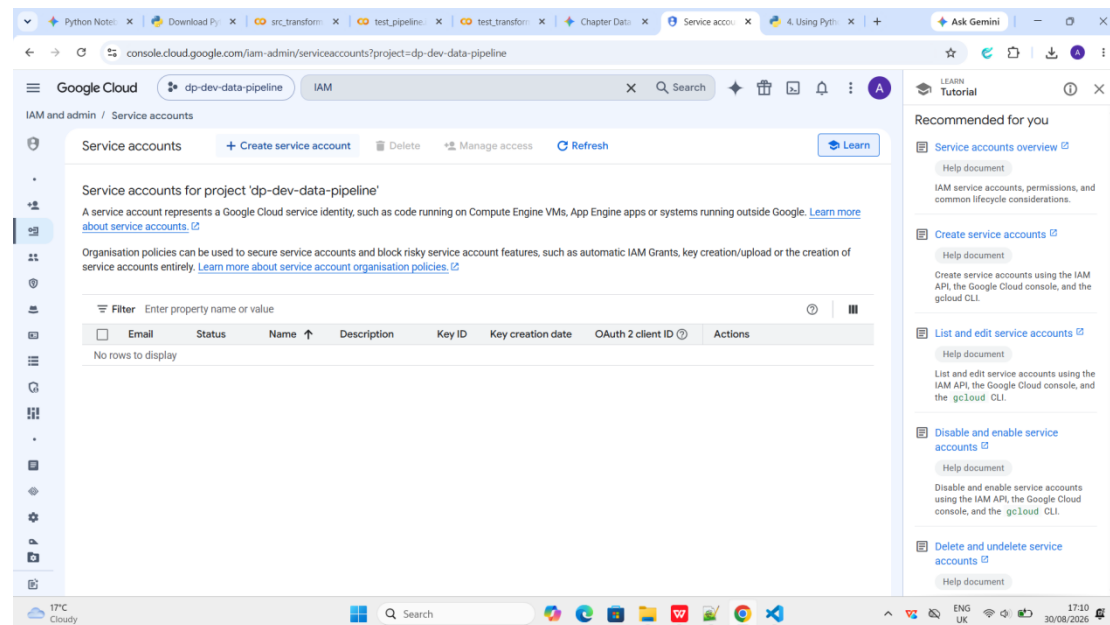
Go to IAM & Admin > Service Accounts



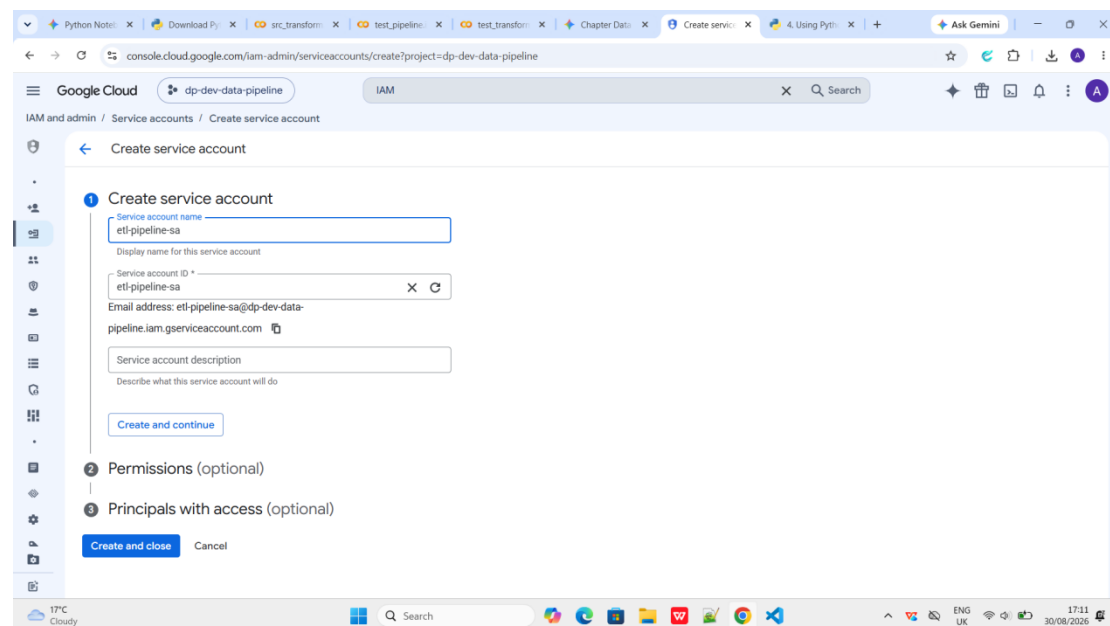
Step 2:- Go to Service Accounts



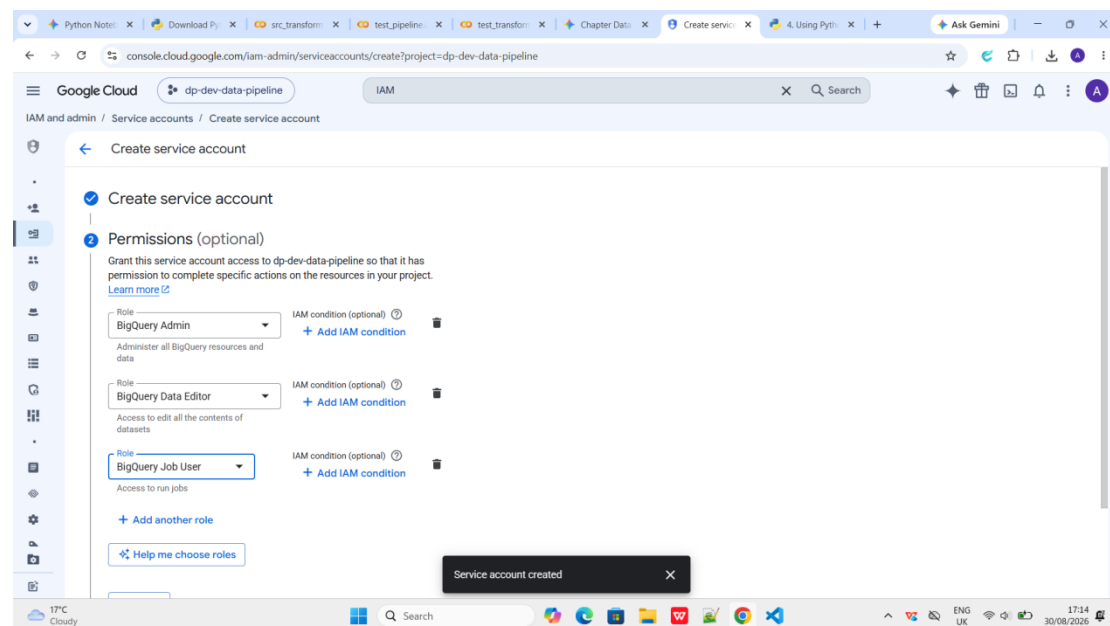
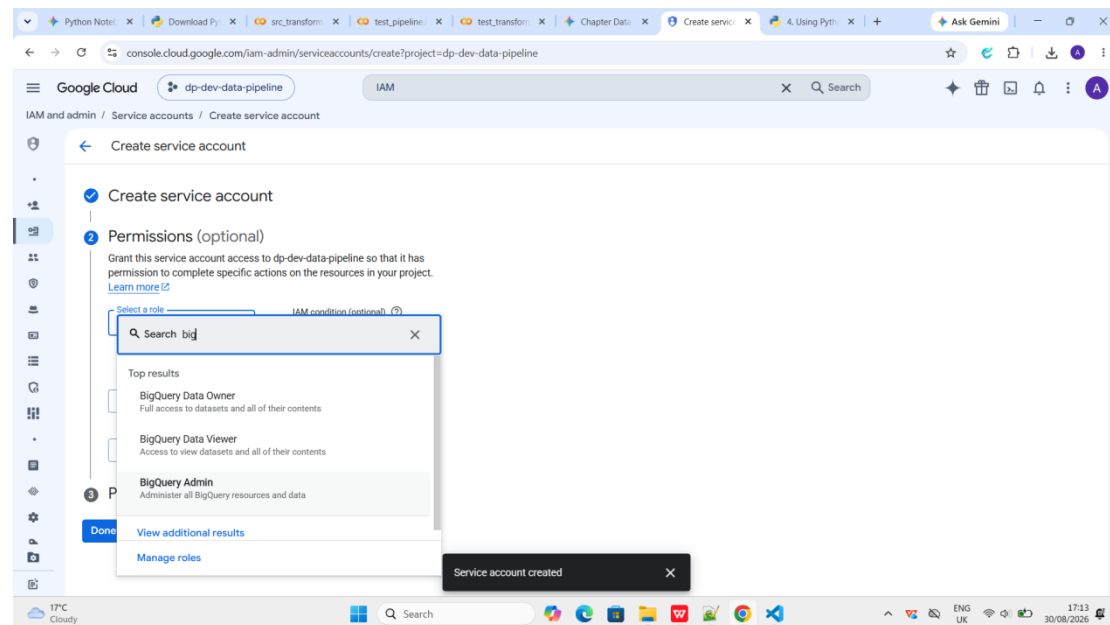
Step 3:- Click on Create Service account at the top



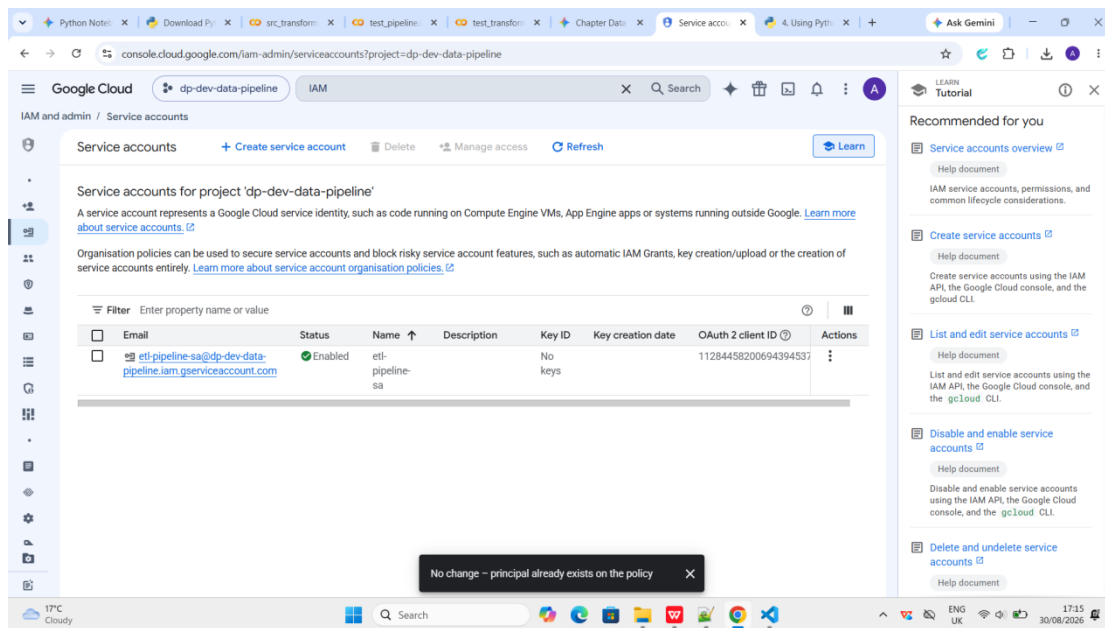
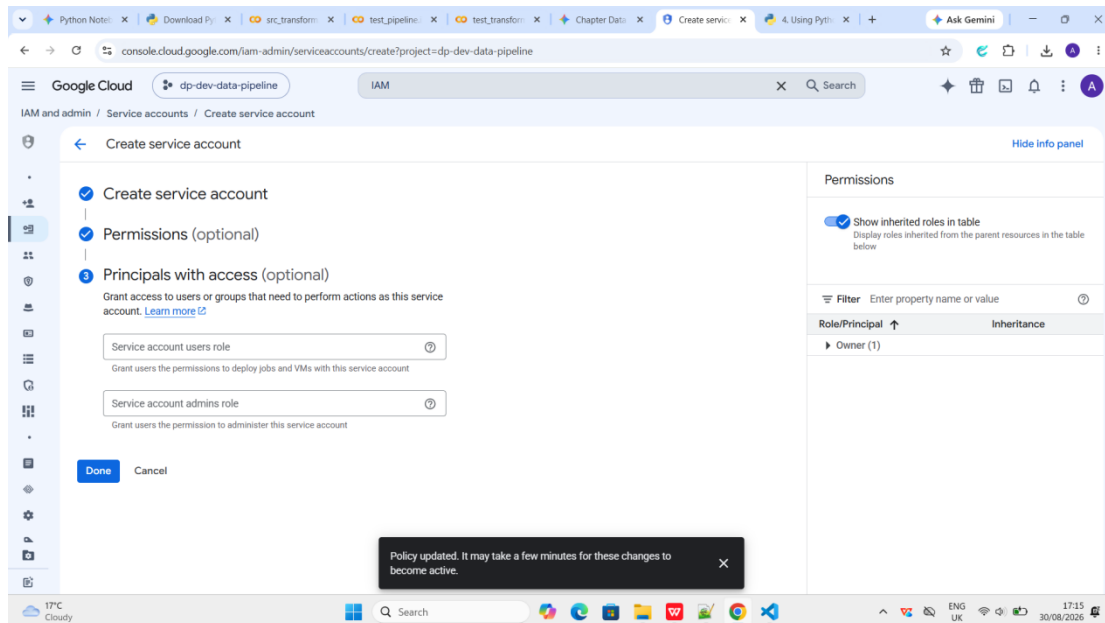
Step 4:- Give name of service account as: etl_pipeline-sa



Step 5:- After that select the roles as BigQuery Admin, BigQuery Data Editor and BigQuery Job User.



Step 6:- Click continue and then Done



Step 7:- Select three dots under the action option and select Manage Keys

Service accounts for project 'dp-dev-data-pipeline'

A service account represents a Google Cloud service identity, such as code running on Compute Engine VMs, App Engine apps or systems running outside Google. [Learn more about service accounts.](#)

Organisation policies can be used to secure service accounts and block risky service account features, such as automatic IAM Grants, key creation/upload or the creation of service accounts entirely. [Learn more about service account organisation policies.](#)

Filter	Email	Status	Name	Description	Key ID	Key creation date	OAuth 2 client ID	Actions
	etl-pipeline-sa@dp-dev-data-pipeline.iam.gserviceaccount.com	Enabled	etl-pipeline-sa		No keys		11284458200694394537	- Manage details - Manage permissions - Manage keys - View metrics - View logs - Disable - Delete

No change - principal already exists on the policy

Step 8:- Click on Add Key and Create new key

etl-pipeline-sa

Details Permissions **Keys** Metrics Logs Principals with access

Keys

Service account keys could pose a security risk if compromised. We recommend that you avoid downloading service account keys and instead use the workload identity federation. [Learn more about the best way to authenticate service accounts on Google Cloud.](#)

Google automatically disables service account keys detected in public repositories. You can customise this behaviour by using the 'iam.serviceAccountKeyExposureResponse' organisation policy. [Learn more](#)

Add a new key pair or upload a public key certificate from an existing key pair.

Block service account key creation using [organisation policies](#).
[Learn more about setting organisation policies for service accounts](#)

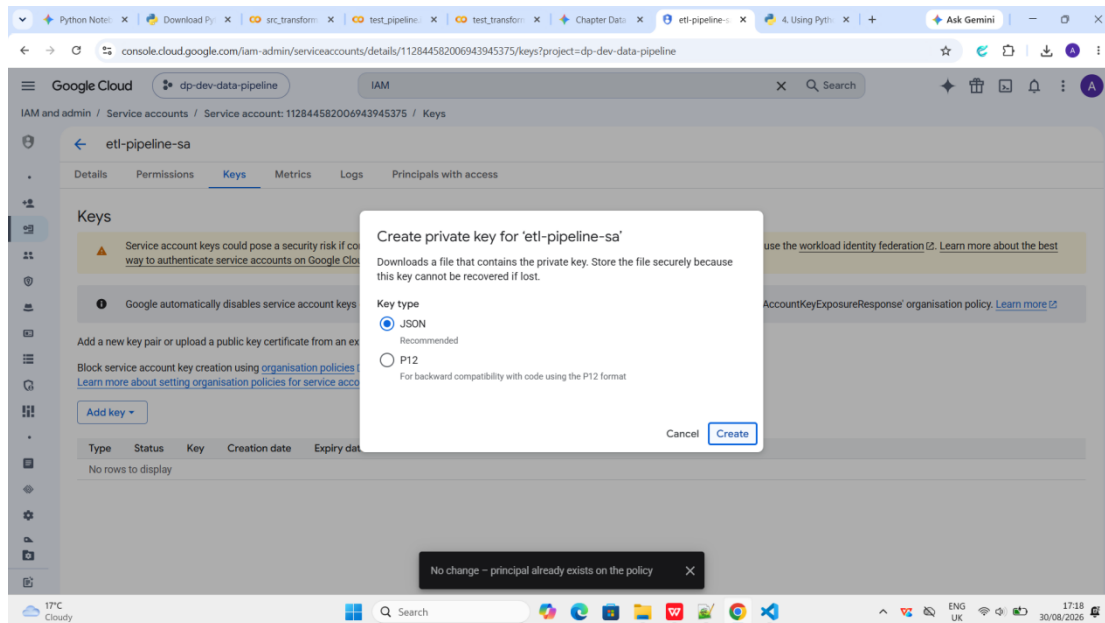
Add key

- Create new key
- Upload existing key

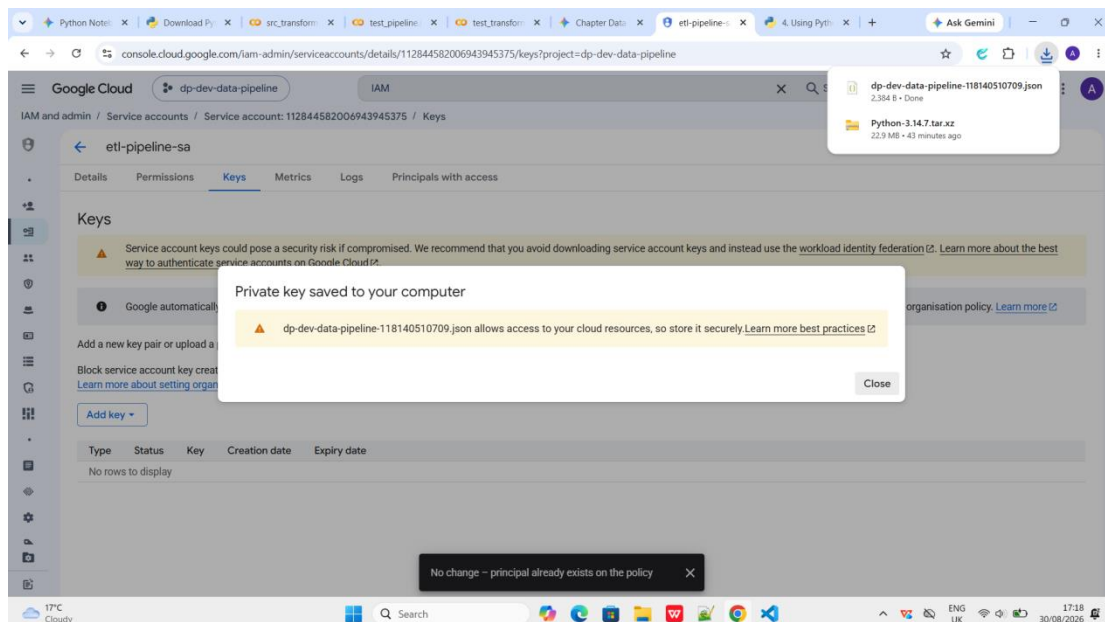
Key	Creation date	Expiry date
-----	---------------	-------------

No change - principal already exists on the policy

Step 9:- Select JSON as the new key



Step 10:- Move that downloaded JSON file into your project folder (for example, inside a dedicated folder or right in your root directory, but make sure it stays private and is never pushed to public repositories like GitHub).



Step 11:- Set the Environment Variable: In your PowerShell terminal, point the environment variable to the exact file path of that downloaded JSON file:

```
$env:GOOGLE_APPLICATION_CREDENTIALS="C:\Users\asipr\Desktop\du-chapters-etl\your-service-account-file.json"
```

Step 12:- Once completed I can run my pipeline:

```
python -m src.src_pipeline
```